

Discussion

In recent years, very user-friendly software has been developed to estimate Bayes factors in (generalized) linear mixed-effects models in the cognitive sciences, including the `bridgesampling`/`brms` packages. However, Bayes factors in complex models have to be estimated, with no guarantees for their accuracy. Here, we used marginal simulation-based calibration for Bayes factors (Schad et al., 2023) in combination with calibration plots (Modrák & Bürkner, in preparation) to test the accuracy of Bayes factor estimation by studying some commonly used factorial experimental designs from the cognitive sciences. The results showed that when no warning message was issued by the bridge sampler, Bayes factors were estimated accurately in a range of well-studied repeated measures (Latin square) designs, which included (a; design 1) a 2×2 design with random effects for subjects, (b; design 2) a design with a single fixed factor and crossed random effects for subjects and items, and (c; design 3) a 2×2 design with crossed random effects for subjects and items. This provides important validation of the relevant Bayes factor estimates. However, in simulations where a warning message was issued by the bridge sampler, we found clear biases in Bayes factor estimates indicating that estimated Bayes factors were mainly too large, and larger than the true Bayes factors, leading to liberal statistical null hypothesis tests that may indicate evidence for an effect even when the effect is in fact zero.

These latter results are encouraging for the application of null hypothesis Bayes factor tests in common factorial designs, which are often used in fields such as psycholinguistics, psychology, and cognitive science. At the same time, the results strongly suggest that warning messages of the bridge sampler need to be taken very seriously. When a warning message was issued by the bridge sampler, the resulting Bayes factor estimates were biased and could not be trusted. They often turned out too large, i.e., providing false support for the alternative hypothesis. Therefore, users of bridge sampling software should be aware that they should not report Bayes factor results when a warning message is shown, and that

outcomes may not only be more variable than usual, but may also be (liberally) biased.

One important question is what a researcher should do in case warning messages are issued by the bridge sampler. Often, those would correspond to computational problems and could be — at least in principle — resolved by a suitable reparametrization or other modification to the model, although this might be difficult in practice. Here, we also studied whether doubling the number of MCMC samples from 10,000 to 20,000 could reduce the number of warning messages. However, we found only a very small improvement. Note also, that we used the `method = "warp3"` argument of the bridge sampler, which may help to reduce warning messages, but warning messages remained. In case this should not be successful either, further options are to report biased Bayes factors together with a discussion of their bias, or to turn to different methods for hypothesis testing. This could involve alternative methods for Bayes factor estimation (such as the Savage-Dickey method or the BayesFactor package) after testing whether they are accurate, but could also involve alternative approaches such as region of practical equivalence (ROPE) (Linde, Tendeiro, Selker, Wagenmakers, & Ravenzwaaij, 2023), or posterior credibility intervals.

Limitations. Overall, our results are limited in that we only focussed on a limited set of analysis settings, priors, models, experimental designs, and software packages. There is thus no guarantee that similar results will be obtained with slightly different designs, models, priors, and data sets. The present results therefore highlight that it is good practice to validate the use of Bayes factors when applying them in practice using simulation-based calibration for Bayes factors (Schad et al., 2023) or alternative approaches (Sekulovski et al., 2024). We hope that the present results will lead to a more robust use and application of Bayes factors by highlighting their strengths and limitations (flagged by warning messages) and also by encouraging researchers to check for their individual analysis, whether estimated Bayes factors are accurate. However, performing SBC for every single analysis can be time consuming and difficult to implement for non-expert users. Therefore, we think that more future research is needed to provide a thorough characterization of the conditions under

which Bayes factor estimates are biased versus accurate.

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Declarations

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- Conflicts of interest/Competing interests

None

- Ethics approval

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- Consent to participate

Not applicable.

- Consent for publication

Not applicable.

- Availability of data and materials

Not applicable.

- Code availability